

Homework Assignment 1: Slab-Geometry Boltzmann Equation

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1 Question 1 - Useful identity for the solution of the time-dependent slab-geometry Boltzmann equation

In this question we are asked to prove the following identity:

$$\psi(x, \mu, t; c) = ce^{-(1-c)t}\psi(cx, \mu, ct; 1) \quad (1)$$

We will do it in two parts. First, we will write down the equation that $\tilde{\psi} = \psi(cx, \mu, ct; 1)$ satisfies, and then we will use it in order to prove that the product $ce^{-(1-c)t}\tilde{\psi}$ satisfies the same equation as $\psi(x, \mu, t; c)$.

1.1 The equation for $\tilde{\psi}$

Notation: We will denote $\frac{\partial \tilde{\psi}(x, \mu, t)}{\partial \tilde{x}}$ as the full derivative (before applying the chain rule) and $\frac{\partial \tilde{\psi}(x, \mu, t)}{\partial x}$ as the partial derivative (after applying the chain rule, and assuming that the cx and ct are parameters *substituted* into the derivative (and not those that the derivative is taken with respect to)). We will denote the same for the time derivative. First, from the definition of $\tilde{\psi}$ we have:

$$\frac{1}{v} \frac{\partial \tilde{\psi}}{\partial \tilde{t}} + \mu \frac{\partial \tilde{\psi}}{\partial \tilde{x}} + \Sigma_t \tilde{\psi} = \frac{1}{2} \int_{-1}^1 \tilde{\psi} d\mu' \quad (2)$$

which after applying the chain rule becomes:

$$\frac{c}{v} \frac{\partial \tilde{\psi}}{\partial t} + c\mu \frac{\partial \tilde{\psi}}{\partial x} + \Sigma_t \tilde{\psi} = \frac{1}{2} \int_{-1}^1 \tilde{\psi} d\mu' \quad (3)$$

1.2 Proving the identity

In this part we are going to simply substitute the RHS of 1 into the LHS of the equation for $\psi(x, \mu, t; c)$ and show that it satisfies the same equation. After substitution and application of the chain rule we get:

$$\text{LHS} = \frac{1}{v} \left[c(c-1)e^{-(1-c)t}\tilde{\psi} + c^2e^{-(1-c)t}\frac{\partial \tilde{\psi}}{\partial t} \right] + \mu c^2e^{-(1-c)t}\frac{\partial \tilde{\psi}}{\partial x} + \Sigma_t ce^{-(1-c)t}\tilde{\psi} \quad (4)$$

$$= ce^{-(1-c)t} \left[\frac{c}{v} \frac{\partial \tilde{\psi}}{\partial t} + c\mu \frac{\partial \tilde{\psi}}{\partial x} + \Sigma_t \tilde{\psi} + \frac{c-1}{v} \tilde{\psi} \right] \quad (5)$$

$$= ce^{-(1-c)t} \left[\frac{1}{2} \int_{-1}^1 \tilde{\psi} d\mu' + \frac{c-1}{v} \tilde{\psi} \right] \quad (6)$$

$$(7)$$

2 Question 2 - Derivation of the time-dependent planar scalar flux for a plane source

The relation between a plane source and a point source as studied in class is given by:

$$\phi_{\text{pl}}(x) = 2\pi \int_{|x|}^{\infty} \phi_{\text{pt}}(r) r dr \quad (8)$$

The solution taken from J. C. J. Paasschens is the point-source scalar flux $\phi_{\text{pt}}(r, t)$ for $c = 1$, and therefore the planar-source scalar flux is given by the integration given above.

$$\phi_{\text{pt}}(r, t) = 2\pi \int_{|x|}^{\infty} \frac{e^{-v\Sigma_t t}}{4\pi r^2} \delta(r - vt) r dr + \int_{|x|}^{\infty} \frac{(1 - r^2/v^2 t^2)^{1/8}}{(4\pi vt/(3\Sigma_t))^{3/2}} e^{-v\Sigma_t t} G\left(v\Sigma_t t \left[1 - \frac{r^2}{v^2 t^2}\right]^{3/4}\right) \Theta(vt - r) r dr \quad (9)$$

$$= \frac{e^{-v\Sigma_t t}}{2vt} \Theta(vt - |x|) + \text{sign}(vt - |x|) \frac{e^{-v\Sigma_t t}}{(4\pi vt/(3\Sigma_t))^{3/2}} \int_{|x|}^{vt} \left(1 - \frac{r^2}{v^2 t^2}\right)^{1/8} G\left(v\Sigma_t t \left[1 - \frac{r^2}{v^2 t^2}\right]^{3/4}\right) r dr \quad (10)$$

$$= \frac{e^{-v\Sigma_t t}}{2vt} \Theta(vt - |x|) + \text{sign}(vt - |x|) \frac{e^{-v\Sigma_t t} (vt)^{1/2}}{(4\pi/(3\Sigma_t))^{3/2}} \int_{\frac{|x|}{vt}}^1 (1 - \xi^2)^{1/8} G\left(v\Sigma_t t [1 - \xi^2]^{3/4}\right) \xi d\xi \quad (11)$$

and then the integral can be evaluated numerically to get the planar-source scalar flux, and of course the relation proved in Q1 can be used to get the solution for $c \neq 1$ using a simple scaling and multiplication by an exponential factor of the solution for $c = 1$.